

ENVARTABELSANALYS DEL 2 SEMINARIUM 22: DIFFERENTIALEKvationer

1 Ordningens differential ekvationer (kap 9.1-9.2)

EX: Lös ekvationen

$$y'(x) + y(x)g(x) = h(x), \text{ där } g(x) \text{ o } h(x) \text{ är givna.}$$

OBSERVERA: Vi har faktorn 1 framför $y'(x)$
som är viktigt att ha

- Sök integrerande faktor (i. f.)

$$- \text{i. f.} = e^{\int g(x) dx} = e^{G(x)}$$

OBS: ingen konstanter, c när du integrerar

- Multiplicera båda leden av ekvationen med den integrerande faktorn:

$$y'(x) \cdot e^{G(x)} + y(x)g(x)e^{G(x)} = h(x)e^{G(x)}$$

Hela vänsterled är:

$$(y(x)e^{G(x)})' = h(x)e^{G(x)}$$

KONTROLL

$$\begin{aligned} (y(x)e^{G(x)})' &= y'(x)e^{G(x)} + y(x)e^{G(x)} \cdot \frac{G'(x)}{g(x)} = \\ &= y'(x)e^{G(x)} + y(x)e^{G(x)}g(x) \end{aligned}$$

$$y(x) \cdot e^{G(x)} = \int h(x)e^{G(x)} dx + C$$

ALLTÄ:

$$y(x) = \frac{1}{e^{G(x)}} \int h(x)e^{G(x)} dx + \frac{1}{e^{G(x)}} + C$$

ELLER

$$y(x) = e^{-G(x)} \int h(x)e^{G(x)} dx + C e^{-G(x)}$$

OBS: GLÖM EJ KONSTANTEN, C

TOLKNING

Lösningarna till detta ek är $f(x)$ ej x .
Alltså vi vill ha en hel funktion,
inte bara ett x .

EX: Lös differential ekvationen

$$xy' + x^2y = x^2, \quad x > 0$$

Lösningar:

$$xg'(x) + x^2g(x) = x^2 \quad / : x, \quad \underline{x > 0}$$

$$1) \quad y'(x) + xg(x) = x = \left[\begin{array}{l} \text{VAR METOD} \\ g(x) = x \\ h(x) = x \end{array} \right]$$

$$2) \quad i.f = e^{\int x dx} = e^{x^2/2}$$

Multiplitera en med $i.f$

$$y'(x) e^{x^2/2} + xg(x) e^{x^2/2} = x e^{x^2/2}$$

3) Nu måste man inse att:

$$(y(x) e^{x^2/2})' = x e^{x^2/2}$$

4) Nya potentes

$$y(x) e^{x^2/2} = \int x e^{x^2/2} dx$$

Lös integralen

$$\int x e^{x^2/2} dx = \left[\begin{array}{l} \frac{x^2}{2} = t \\ x^2 = 2t \\ 2x dx = 2dt \\ x dx = dt \end{array} \right] = \int e^t dt = e^t + C = e^{x^2/2} + C$$

5 TILLBAKA TILL EKVATIONEN

$$y(x) e^{x^2/2} = e^{x^2/2} + C \quad / : e^{-x^2/2} \quad \text{kan även dela med}$$

$$y(x) \frac{e^{x^2/2} \cdot e^{-x^2/2}}{e^0 = 1} = \frac{e^{x^2/2} \cdot e^{-x^2/2} + C e^{-x^2/2}}{e^0 = 1}$$

$$y(x) = 1 + C e^{-x^2/2}$$

SVAR: $y(x) = 1 + C e^{-x^2/2}, \quad x > 0$

OBS: Kontrollera
gör att utföra
Bör göras

KONTROLL:

$$VL = xy' + x^2y = [y = 1 + Ce^{-x^2/2}] =$$

$$= x(Ce^{-x^2/2}(-x)) + x^2(1 + Ce^{-x^2/2})$$

$$= \cancel{-x^2(Ce^{-x^2/2})} + x^2 + \cancel{x^2Ce^{-x^2/2}} = x^2$$

$$HL = x^2 \quad \text{och} \quad VL = x^2$$

Så korrekt, $y(x) = 1 + Ce^{-x^2/2}$ är en lösning

Ex: LÖS

$$\begin{cases} y'(x) = x - y(x) \\ y(0) = 0 \end{cases}$$

• LÖSNING:

$$y'(x) = x - y(x) \Leftrightarrow y'(x) + y(x) = x$$

$$= [g(x) = 1 \quad h(x) = x]$$

(1) SÖK integrerande faktor

$$i.f. = e^{\int 1 dx} = e^x$$

(2) Multipluera hela ekvationen med i.f.

$$y'(x)e^x + y(x)e^x = xe^x$$

$$(y(x)e^x)' = xe^x$$

$$y(x)e^x = \int xe^x dx$$

$$\begin{aligned} \bullet \text{ IIL} &= \int xe^x dx = \left[\begin{array}{l} u = x \quad u' = 1 \\ v = e^x \quad v' = e^x \end{array} \right] \\ &= xe^x - \int e^x dx = xe^x - e^x + C \end{aligned}$$

3) TILLBAKA TILL EKVATIONEN

$$y(x)e^x = xe^x - e^x + C \quad | \cdot e^{-x}$$

$$y(x) \underbrace{e^x e^{-x}}_{e^0=1} = x \underbrace{e^x e^{-x}}_{e^0=1} - \underbrace{e^x e^{-x}}_{e^0=1} + C e^{-x} =$$

$$y(x) = x - 1 + C e^{-x} \quad (\text{s.k Allmän lösning})$$

4) BIVILLKOR:

Villkoret $y(0) = 0$ ger

$$y(0) = 0 - 1 + e^0 C = -1 + C = 0$$

$$-1 + C = 0 \Leftrightarrow C = 1$$

$$C = 1 \text{ ger LÖSNINGEN: } y(x) = x - 1 + 1 \cdot e^{-x}$$

$$\text{SVAR: } x - 1 + e^{-x}$$

Ex: Finn den lösning till diff ek

$(1+x^2)y' + y = 2$, som har gränsvärdet 3 då $x \rightarrow +\infty$

Lösning:

$$\begin{cases} (1+x^2)y' + y = 2 \\ \lim_{x \rightarrow \infty} y(x) = 3 \end{cases} \quad \left| \begin{array}{l} (1+x^2)y'(x) + y(x) = 2 \quad / : (1+x^2) \\ y'(x) + \frac{1}{1+x^2} y(x) = 2 \cdot \frac{1}{1+x^2} \\ \left[g(x) = \frac{1}{1+x^2} \quad \text{och} \quad h(x) = \frac{2}{1+x^2} \right] \end{array} \right.$$

Integrerande faktor:

$$e^{\int \frac{1}{1+x^2} dx} = e^{\arctan(x)}$$

Multiplitera en med i.f
 $y'(x) \cdot e^{\arctan(x)} + \frac{1}{1+x^2} y(x) e^{\arctan(x)} = \frac{2}{1+x^2} e^{\arctan(x)}$

Skriv nu om

$$(y(x) e^{\arctan(x)})' = \frac{2}{1+x^2} e^{\arctan(x)}$$

$$y(x) e^{\arctan(x)} = \int \frac{2}{1+x^2} e^{\arctan(x)} dx$$

$$\begin{aligned} \text{HL: } 2 \int \frac{1}{1+x^2} e^{\arctan(x)} dx &= \left[\arctan(x) = t \right] \\ &= 2 \int e^t dt = 2e^t + C \\ &= 2e^{\arctan(x)} + C \end{aligned}$$

TILLBAKA TILL EKVATIONEN

$$y(x) e^{\arctan(x)} = 2e^{\arctan(x)} + C \quad / : e^{-\arctan(x)}$$

$$y(x) = 2 + C e^{-\arctan(x)} \quad (\text{Allmän lösning})$$

ANVÄND VILLKORET

$$\begin{aligned} \lim_{x \rightarrow \infty} y(x) &= 3, \text{ ger } \lim_{x \rightarrow \infty} (2 + C e^{-\arctan(x)}) = 3 \\ 2 + C e^{-\pi/2} &= 3 \Rightarrow C e^{-\pi/2} = 1 \Rightarrow C e^{-\pi/2} \cdot e^{\pi/2} = e^{\pi/2} \\ C &= e^{\pi/2} \end{aligned}$$

SVAR: $2 + e^{\pi/2} e^{-\arctan(x)} = y(x)$

Ex: Lös diff en

$$\cos(x) y'(x) = 2(\sin(x))y + 2(\cos(x)+1)$$

$$0 < x < \frac{\pi}{2}$$

→ Detta är så $\sin$ & $\cos$ är pos

Lösning:

$$(\cos x) y' = 2(\sin x) y + 2(\cos x + 1) \quad | : \cos x$$

$\cos x \neq 0$
ty $0 < x < \pi/2$

$$y' - \frac{2 \sin(x)}{\cos(x)} y = \frac{2(\cos x + 1)}{\cos x}$$

$$\left[g(x) = -\frac{2 \sin(x)}{\cos(x)} \quad \text{och} \quad h(x) = \frac{2(\cos(x)+1)}{\cos(x)} \right]$$

INTEGRERANDE FAKTOR

$$e^{\int \left(\frac{-2 \sin(x)}{\cos(x)} \right) dx} = 2 \int \frac{-\sin(x)}{\cos(x)} dx = 2 \ln |\cos(x)|$$

$$e^{2 \ln |\cos(x)|} = \cos(x), \quad 0 < x < \pi/2$$
$$\Rightarrow |\cos(x)| = \cos(x)$$
$$= e^{2 \ln(\cos(x))} = e^{\ln(\cos(x))^2} = (\cos(x))^2$$

MULTIPLICERA med if

OBS!

$$y' \cos^2(x) - \frac{2 \sin(x)}{\cos(x)} y \cos^2(x) = \frac{2(\cos(x)+1) \cos^2(x)}{\cos(x)}$$

$$- y' \cos^2(x) - 2 \sin(x) y \cos(x) = 2(\cos(x)+1) \cos(x)$$

$$(y \cos^2(x))' = \int 2(\cos(x)+1) \cos(x) dx$$

$$y \cos^2(x) = \int (2 \cos^2(x) + 2 \cos(x)) dx$$

$$H_1 = \left[\cos(2x) = 2 \cos^2(x) - 1 \Rightarrow \frac{\cos(2x)+1}{2} \right]$$
$$= \frac{1}{2} (\cos(2x)+1)$$

$$= \int \left(2 \cdot \frac{1}{2} (\cos(2x)+1) + 2 \cos(x) \right) dx$$

$$= \int (\cos(2x) + 1 + 2 \cos(x)) dx =$$

$$= \int \cos(2x) dx + \int 1 dx + 2 \int \cos(x) dx =$$

$$\left[\begin{array}{l} 2x = t \\ \text{anws} \end{array} \right] = \frac{\sin(2x)}{2} + x + 2\sin(x) + C$$

• Tillbaka till ekvationen

$$y(x) \cos^2(x) = \frac{1}{2} \sin(2x) + x + 2\sin(x) + C \quad / : (\cos^2(x))$$

$$y(x) = \frac{1}{2} \frac{\sin(2x)}{\cos^2(x)} + \frac{x}{\cos^2(x)} + \frac{2\sin(x)}{\cos^2(x)} + \frac{C}{\cos^2(x)}$$

SVAR:

$$y(x) = \frac{1}{2} \cdot \frac{\sin(2x)}{\cos^2(x)} + \frac{x}{\cos^2(x)} + 2 \frac{\sin(x)}{\cos^2(x)} + \frac{C}{\cos^2(x)} + \frac{C}{\cos^2(x)}$$

OBS

$$\frac{\sin(2x)}{\cos^2(x)} = \frac{2\sin(x) \cancel{\cos(x)}}{\cancel{\cos(x)} \cos^2(x)} = 2 \frac{\sin(x)}{\cos(x)} = 2 \tan(x)$$